Engineering Principles: Where Logic Should Live

1. Code is the Source of Truth

All business logic must live in code. Infrastructure should only execute, not decide.

2. Control Over System Behavior

Code provides deterministic and predictable behavior. Infrastructure logic creates hidden side effects.

3. Testability First

If logic cannot be unit/integration tested easily, it is in the wrong place.

4. Versioning and Change Management

Code is versioned, diffable, and rollback-safe. Infrastructure logic often is not.

5. Reproducibility

A system should behave the same when reproduced locally or in CI/CD.

6. Debuggability

Code allows step-by-step debugging. Infrastructure logic forces guesswork.

7. Typed Contracts Over Dynamic Blobs

Strong typing prevents runtime surprises and enforces correctness.

8. Complexity Must Be Managed, Not Hidden

Moving logic to tools (RabbitMQ, n8n, configs) does not reduce complexity—it hides it.

9. Separation of Concerns

Broker = transport layer

Services = business logic

10. Golden Rule

"If it contains decisions — it belongs to code."

11. Acceptable Infrastructure Responsibilities

- Routing (basic)
- Delivery guarantees
- Retries (DLX)
- Security

12. Anti-patterns

- Business logic in brokers
- Message transformations in infrastructure
- Hidden orchestration in tools

13. Long-term Thinking

Optimize for maintainability over short-term speed.

14. Mental Model

Code → source of truth

Infrastructure → execution environment